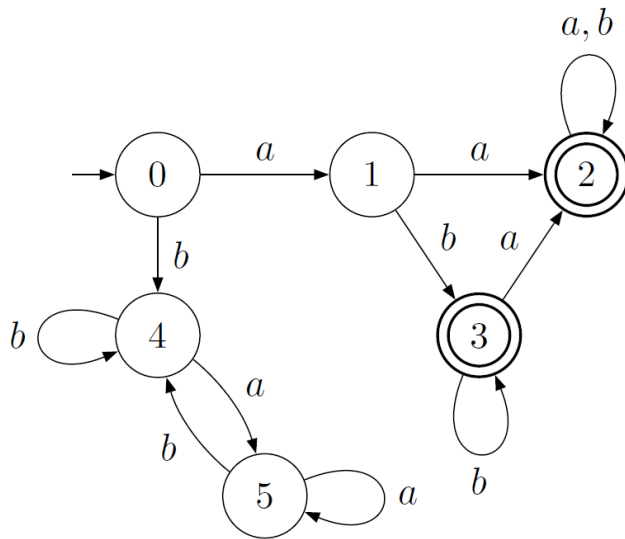




G52LAC (COMP2040)

Languages and Computation

Lecture 2: Alphabets, Words and Languages

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Outline

- Quick recap. of the previous lecture.
- Quick revision: sets, ordered tuples, functions and relations.
- Terminology.
- Symbols & alphabets.
- Words.
- Languages.
- Revisiting the Chomsky Hierarchy.

Sets vs. ordered n-tuples

- Ordered n-tuples vs. sets.
- Ordered n-tuples:
 - Order matters.
 - Elements can repeat.
 - E.g. $(3, 2, 2, 5, 9, 5)$.
- Sets:
 - Order does not matter.
 - Elements can't repeat.
 - E.g. $\{2, 3, 5, 9\}$.



Relation?
Function?
Comparison?

Functions and Relations

- **Cartesian product** between two sets A and B:
 - The set of all ordered 2-tuples (a,b) where a belongs to A and b belongs to B.
 - Whiteboard: $A = \{1,2\}$ and $B = \{5,6,7\}$...
- A relation is a **generalization** of a function.
- Consider two sets A and B.
 - $A \times B$. **Cartesian product** between A and B.
 - Any subset of $A \times B$ is a relation between A and B.
 - A function is a **special** relation R between A and B:
 - Whenever (a,b) and (a,b') are in the relation R, then $b = b'$.

Terminology

- The terms **alphabet**, **word** and **language** are used in a strict technical sense in this course.
- An **alphabet** is a **finite set** of symbols.
- A **word** is a **finite sequence** of symbols.
- A **language** is a **set** of words.
- Languages can be finite or infinite.
- The term **string** is often used interchangeably with the term **word**.

Symbols & Alphabets

- What is a symbol, then?
- Anything, but it has to come from an alphabet.
- Usually, Σ is used to denote an alphabet.
- Example alphabets:

$$\Sigma_1 = \{0, 1\}$$

$$\Sigma_2 = \{a, b, c, d, e, f, g, h, i, j, k, l, m, \\ n, o, p, q, r, s, t, u, v, w, x, y, z\}$$

$$\Sigma_3 = \{\circ, \square, \triangle\}$$

$$\Sigma_4 = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, +, -, *, /\}$$

- Important exception: ε is never used as an alphabet symbol.

The Empty Word

- ε is used to denote the **empty word**: the sequence of zero symbols.
- But ε itself is not a symbol!
- ε is a **word**, not a set.
- So don't confuse it with the **empty set** (denoted $\emptyset$ or $\{\}$).
- Thus, $\{\varepsilon\} \neq \{\}$.

Words over an alphabet

- The set of **all** words over an alphabet Σ is denoted by Σ^* .
- Σ^* can be defined inductively as follows:
 - $\varepsilon \in \Sigma^*$ Inductive or recursive definition
 - if $x \in \Sigma$ and $w \in \Sigma^*$ then $xw \in \Sigma^*$
- Note that $\varepsilon \in \Sigma^*$ for **any** alphabet Σ (including $\Sigma = \emptyset$).
- Iff $\Sigma \neq \emptyset$ then Σ^* is an **infinite** set (of **finite** words).
So when can Σ^* be finite?

Note 1: a word is essentially an ordered n-tuple without punctuation, e.g. the word **cat** corresponds to the ordered 3-tuple **(c,a,t)**.

Note 2: The **inductive definition of a set S** consists of first specifying one or more elements of the set S and then specifying one or more construction rules to obtain other elements of S based on existing elements of S.

Example

- Given $\Sigma = \{0, 1\}$, some elements of Σ^* are:



Recall:

- The set of **all** words over an alphabet Σ is denoted by Σ^* .
- Σ^* can be defined inductively as follows:
 - $\varepsilon \in \Sigma^*$
 - if $x \in \Sigma$ and $w \in \Sigma^*$ then $xw \in \Sigma^*$
- Note that $\varepsilon \in \Sigma^*$ for **any** alphabet Σ (including $\Sigma = \emptyset$).
- Iff $\Sigma \neq \emptyset$ then Σ^* is an **infinite** set (of **finite** words).

Example

- Given $\Sigma = \{0, 1\}$, some elements of Σ^* are:

ε ,

0, 1,

00, 10, 01, 11,

000, 100, 010, 110, 001, 101, 011, 111,

0000, ...

- This is just applying the inductive definition.
- Important note: only write ε if it appears on its own, as it denotes an **absence** of symbols.

e.g. 000ε is redundant; you can just depict the word as 000 .

Alternative Notation

- The set of all words over Σ of length n is denoted by Σ^n (where $n \in \mathbb{N}$).
- For example, if $\Sigma = \{a, b\}$, then $\Sigma^2 = \{aa, ab, ba, bb\}$.
- This can be used to give an alternative (but equivalent) definition of Σ^* :

$$\Sigma^* = \bigcup_{n=0}^{\infty} \Sigma^n$$

- Remember that in computer science, $0 \in \mathbb{N}$.

Languages

- A language L over an alphabet Σ is a subset of Σ^* :

$$L \subseteq \Sigma^*$$

or

$$L \in \mathcal{P}(\Sigma^*)$$



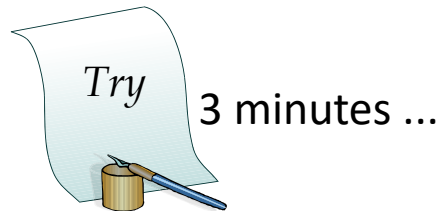
Remember “power set”?

- A language may be a **finite** or **infinite** set.
- Note that while ε is always an element of Σ^* , it may or may not be an element of an arbitrary language.

Exercise

Given $\Sigma = \{a, b, c\}$, define some languages over Σ .

Try to define some **finite** ones and some **infinite** ones.



Exercise

Given $\Sigma = \{a, b, c\}$, define some languages over Σ .

- $\{a, abba, baa, cab\}$
- $\{c\}$
- $\{\varepsilon, a, bbb\}$
- $\{\varepsilon\}$
- $\{a^n \mid n \in \mathbb{N}\}$
- $\{a^n b^n \mid n \in \mathbb{N}, n \geq 10\}$
- $\{w \mid w \in \Sigma^*, \text{odd}(\text{length}(w))\}$
- $\emptyset$
- Σ^*

Concatenation of Words

- An important operation on words is **concatenation**.
- Concatenation is denoted by **juxtaposition** (i.e. writing the words side by side without using an operator symbol).
- If $v \in \Sigma^*$ and $w \in \Sigma^*$ then $vw \in \Sigma^*$
- Concatenation can be defined by primitive recursion:

$$\begin{aligned}\varepsilon w &= w \\ (xv)w &= x(vw)\end{aligned}$$

where

$$\begin{aligned}x &\in \Sigma \\ v, w &\in \Sigma^*\end{aligned}$$

Properties of Word Concatenation

- Concatenation is **associative** and has **unit** ε :

$$u (vw) = (uv) w$$

$$\varepsilon u = u = u \varepsilon$$

where

$$u, v, w \in \Sigma^*$$

- Concatenation of words is **not commutative** (i.e. order matters), as words are sequences.

$$vw \neq wv$$

Concatenation of Languages

- Remember, languages are **sets**, not sequences.
- Given two **languages** M and N over an alphabet Σ , their concatenation (MN) is defined:

$$MN = \{uv \mid u \in M \wedge v \in N\}$$

- Example:

$$\Sigma = \{a, b, c\}$$

$$M = \{\varepsilon, a, aa\}$$

$$N = \{b, c\}$$

$$MN = \{uv \mid u \in \{\varepsilon, a, aa\} \wedge v \in \{b, c\}\}$$

$$= \{\varepsilon b, \varepsilon c, ab, ac, aab, aac\}$$

$$= \{b, c, ab, ac, aab, aac\}$$

Properties of Language Concatenation (1)

- Concatenation of languages is **associative**:

$$L (MN) = (LM) N$$

- Concatenation of languages has **zero** $\emptyset$ (the empty language):

$$L\emptyset = \emptyset = \emptyset L$$

- Concatenation of languages has **unit** $\{\varepsilon\}$ (the language containing only the empty word):

$$L \{\varepsilon\} = L = \{\varepsilon\} L$$

Properties of Language Concatenation (2)

- Concatenation of languages distributes through set union:

$$\begin{aligned}L(M \cup N) &= LM \cup LN \\(L \cup M)N &= LN \cup MN\end{aligned}$$

- But it **does not** distribute through set intersection:

$$L(M \cap N) \neq LM \cap LN$$

- Counterexample:

$$\begin{aligned}L &= \{\varepsilon, a\}, M = \{\varepsilon\}, N = \{a\} \\L(M \cap N) &= L\emptyset = \emptyset \\LM \cap LN &= \{\varepsilon, a\} \cap \{a, aa\} = \{a\}\end{aligned}$$

Concatenating a Language with Itself

- A language can be concatenated with itself.
- Exponent notation is often used for this:
 - $L^1 = L$
 - $L^2 = LL$
 - $L^3 = LLL$
 - $L^4 = LLLL$
 - etc. . .
- L^0 is defined to be $\{\varepsilon\}$.
(As $\{\varepsilon\}$ is the unit of concatenation.)

Kleene Star

- Given $L \subseteq \Sigma^*$, L^* is zero or more concatenations of L .
- Note that these are different stars (but both mean 'zero or more').

$$L^* = \{w_0 w_1 \dots w_{n-1} \mid n, i \in \mathbb{N}, \forall i < n, w_i \in L\}$$

or

$$L^* = \bigcup_{n=0}^{\infty} L^n = L^0 \cup L^1 \cup L^2 \cup \dots$$

or

$$\begin{aligned} \varepsilon &\in L^* \\ w \in L &\Rightarrow w \in L^* \\ v \in L^* \wedge w \in L^* &\Rightarrow vw \in L^* \end{aligned}$$

Language Membership

- Fundamental question of this module:

Given a language $L \subseteq \Sigma^$ and a word $w \in \Sigma^*$, can we determine if $w \in L$?*

- If L is finite, this is easy.
- But not so easy if L is infinite, which most interesting languages are.
- We need:
 - A **finite** (and preferably concise) **description** of the (infinite) language.
 - A method to **decide** if $w \in L$ or not, given such a description.
- Over the course of this module we are going to encounter a number of possibilities, with varying descriptive power.

Recommended Reading

- Introduction to Automata Theory, Languages, and Computation (3rd edition), pages 28–33.
- Original G52MAL Lecture Notes, page 6.

Easy Exercises

- An **alphabet** is a _____ of symbols.
- A **word** is a _____ of symbols.
- A **language** is a _____ of words.
- Can languages be infinite?

Easy Exercises

- What is ε ?
- Is it a set? Is it the same as $\{ \}$?

Easy Exercises

- Given an alphabet Σ , explain in your own words the meaning of Σ^* .
- Given $\Sigma = \{a,b,c\}$, list five elements from Σ^* .

Easy Exercises

- Consider the languages $L1 = \{1,22\}$ and $L2 = \{a,bb\}$.
- List down the elements of $L3$, which is defined as the concatenation $L1 L2$.

Question from a previous exam

(a) [A] Consider the following languages:

[4 marks]

$$L1 = \{a,b,c\}$$

$$L2 = \{c,d,e\}$$

$$L3 = \{c,d,f\}$$

$$L4 = \{g,h\}$$

List all words belonging to the following languages:

[1 mark each]

$$L5 = (L1 \cup L2) \cap L3$$

$$L6 = (L1 \cap L2)(L2 \cap L3)$$

$$L7 = (\{\varepsilon\}L1\{\varepsilon\}) \cap L3$$

$$L8 = (L1 \cup L3)(L3 \cap L4)$$

Question from a previous exam

(a) [A] Consider the following languages:

[4 marks]

$$L1 = \{a,b,c\}$$

$$L2 = \{c,d,e\}$$

$$L3 = \{c,d,f\}$$

$$L4 = \{g,h\}$$

List all words belonging to the following languages:

[1 mark each]

$$L5 = (L1 \cup L2) \cap L3 = \{c,d\}$$

$$L6 = (L1 \cap L2)(L2 \cap L3) = \{cc,cd\}$$

$$L7 = (\{\varepsilon\}L1\{\varepsilon\}) \cap L3 = \{c\}$$

$$L8 = (L1 \cup L3)(L3 \cap L4) = (L1 \cup L3)\emptyset = \{\}$$

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